Answer-1

Covariance measures **how two variables vary together**.

For variables X and Y:

$$\text{Cov}(X, Y) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

- **Positive covariance:** when X is above its mean, Y also tends to be above its mean (move in same direction).
- **Negative covariance:** when X is above its mean, Y tends to be below its mean (move in opposite directions).

1. Scale

- **Covariance is scale-dependent**
  - Its value depends on the **units** of X and Y.
  - If you change units (e.g., cm to m, rupees to dollars), the covariance value changes.
- **Correlation is scale-free**
  - It is the **standardized** form of covariance:

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Always lies between -1 and $+1$, independent of units.

2. Interpretation

- **Covariance:**
  - Only tells **direction** (positive/negative).
  - Magnitude is **hard to interpret** because it has no fixed range and depends on units.
- **Correlation:**
  - Tells both **direction** and **strength** of linear relationship.
  - Values close to ± 1= strong, close to 0 = weak.
  - It is easy to compare across different datasets and variables.

Answer-2

<u>Positive Covariance</u>

- When one variable **increases**, the other also **increases**.

- When one **decreases**, the other also **decreases**.
- They move **in the same direction**.

Example: Experience ↑ → Salary ↑ (generally)

<u>Negative Covariance</u>

- When one variable **increases**, the other **decreases**.
- When one **decreases**, the other **increases**.
- They move **in opposite directions**.

Example: Price ↑ → Quantity demanded ↓

<u>Zero Covariance</u>

- No linear relationship between the variables.
- The variables do **not move together** in any consistent pattern.
- They may still have a **non-linear** relationship, but covariance won't capture it.

Answer-3

<u>Limitations of Covariance</u>

1. Covariance is scale-dependent

Its value changes if you change the units of measurement. Example:

- **Salary in rupees vs dollars**
- **Height in cm vs meters the covariance value will change even though the relationship stays the same.**

2. Magnitude is not interpretable

A covariance of 2000 — is that strong or weak? You cannot tell without knowing the units and variability of both variables.

3.No fixed range

Covariance can take any value from negative infinity to positive infinity. This makes it impossible to compare relationships across different datasets.

4.Only shows direction, not strength

Covariance tells you:

- **Positive → move together**
- **Negative → move opposite but it does not tell how strong the relationship is.**

<u>Why correlation is preferred</u>

Unit-free and scale-free

Correlation standardizes covariance by dividing by standard deviations. This removes the effect of units.

Bounded between –1 and +1

This fixed range makes interpretation easy:

- **+1 perfectly positive**
- **–1 → perfect negative**
- **0 → no linear relationship**

Comparable across datasets

Because correlation is standardized, you can compare relationships even if variables have different units or scales.

Shows both direction and strength

Correlation gives a complete picture of the linear relationship.

Answer-4

<u>Pearson's Correlation Coefficient (r)</u>

- **Measures the strength and direction of a *linear* relationship between two variables.**
- **Uses actual numerical values.**
- **Assumes:**
    - **Data is continuous**
    - **Relationship is linear**
    - **Variables are normally distributed**
    - **No extreme outliers**

Best for: Linear, continuous, normally distributed data.

<u>Spearman's Rank Correlation Coefficient (ρ or Rs)</u>

- **Measures the strength and direction of a *monotonic* relationship (not necessarily linear).**
- **Uses ranks, not raw values.**
- **Does not require normality.**
- **Resistant to outliers because it uses ranks.**
- **Works with:**
    - **Ordinal data**
    - **Non-linear but monotonic relationships**

Best for:

- **Data that is not normally distributed**
- **Data with outliers**
- **Ordinal variables**
- **Non-linear but monotonic relationships**

<u>Use Spearman's when</u>:

- **The relationship is monotonic but not linear**
- **Data contains outliers**
- **Variables are ordinal (ranked)**
- **Data violates Pearson's assumptions (non-normal, skewed, uneven spacing)**

Example: Ranking students by performance, customer satisfaction ratings, or any situation where the *order* matters more than the exact value.

Answer-5

Interpretation of r = 0.85

- **A correlation of 0.85 indicates a strong positive linear relationship between variables X and Y.**
- **This means:**
    - **When X increases, Y also tends to increase.**
    - **The data points lie close to an upward-sloping straight line.**
    - **X is a good predictor of Y in this dataset.**

So, the relationship is strong, positive, and linear.

No — correlation does NOT imply causation.

Even though the correlation is strong, it does not prove that:

- **X causes Y**
- **Y causes X**
- **Or that the relationship is direct**

Why not?

1. **A third variable may be influencing both Example: income might affect both education and spending.**
2. **Correlation does not show direction We cannot tell whether X → Y or Y → X.**
3. **Could be coincidence or due to external factors Strong correlations can occur without any causal link.**

Answer-6

<u>Mean values</u>

$$\bar{X} = \frac{2+4+6+8}{4} = 5$$

$$\bar{Y} = \frac{3+7+5+10}{4} = 6.25$$

<u>Compute $(X-\bar{X})(Y-\bar{Y})$</u>

X	Y	$X-\bar{X}$	$Y-\bar{Y}$	Product
2	3	-3	-3.25	9.75
4	7	-1	0.75	-0.75
6	5	1	-1.25	-1.25
8	10	3	3.75	11.25

$$\sum(X-\bar{X})(Y-\bar{Y}) = 19$$

<u>Covariance (sample)</u>

$$\text{Cov}(X,Y) = \frac{19}{4-1} = \boxed{6.33}$$

Answer-7

<u>Means</u>

$$\bar{A} = 30, \bar{B} = 18.4$$

<u>Compute correlation</u>

$$r = \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \sum(B - \bar{B})^2}}$$

<u>After calculation</u>:

$$r \approx \boxed{0.997}$$

Interpretation: Very strong positive correlation.

Answer-8

<u>Mean values</u>

$$\bar{H} = 165, \bar{W} = 59$$

<u>Pearson correlation</u>

$$r = \frac{\sum(H - \bar{H})(W - \bar{W})}{\sqrt{\sum(H - \bar{H})^2 \sum(W - \bar{W})^2}}$$

<u>After calculation</u>:

$$\boxed{r \approx 0.98}$$

Conclusion: Strong positive correlation between height and weight.

Answer-9

X 1 2 3 4 5
Y 15 12 9 7 3

As X increases, Y decreases.

Negative correlation

Answer-10

Mean Returns

$$\bar{A} = \frac{8 + 10 + 12 + 9 + 11}{5} = 10$$

$$\bar{B} = \frac{6 + 9 + 11 + 8 + 10}{5} = 8.8$$

Covariance Calculation

A	B	$A - \bar{A}$	$B - \bar{B}$	Product
8	6	-2	-2.8	5.6
10	9	0	0.2	0
12	11	2	2.2	4.4
9	8	-1	-0.8	0.8
11	10	1	1.2	1.2

$$\sum(A - \bar{A})(B - \bar{B}) = 12$$

Sample Covariance

$$\text{Cov}(A, B) = \frac{12}{5 - 1} = \boxed{3}$$

Correlation Coefficient

$$r = \frac{\sum(A - \bar{A})(B - \bar{B})}{\sqrt{\sum(A - \bar{A})^2 \sum(B - \bar{B})^2}}$$

$$r = \frac{12}{\sqrt{10 \times 14.8}} \approx \boxed{0.99}$$

<u>Interpretation</u>

- **Covariance = +3 → Portfolios move in the same direction**
- **Correlation ≈ 0.99 → Very strong positive relationship**

Conclusion

The two portfolios move together very closely.